

Problem

The voltage $v(t)$ across a device and the current $i(t)$ through it are

$$v(t) = 20 \sin(4t) \text{ V and } i(t) = 10(1 + e^{-2t}) \text{ mA}$$

Calculate:

(a) the total charge in the device at $t = 1 \text{ s}$, $q(0) = 0$.

(b) the power consumed by the device at $t = 1 \text{ s}$.

Step-by-step solution

Step 1 of 2

(a)

The voltage across a given device is $v(t) = 20 \sin(4t) \text{ V}$ and the current through it is $i(t) = 10(1 + e^{-2t}) \text{ mA}$. Given $q(0) = 0$ and the total charge in the device at $t = 1 \text{ s}$ is,

$$\begin{aligned} q &= \int i dt \\ &= \int_0^1 10 \times 10^{-3} (1 + e^{-2t}) dt \\ &= 10 \times 10^{-3} \left(t - \frac{e^{-2t}}{2} \right)_0^1 \\ &= 10 \times 10^{-3} \left((1 - 0) - \left(\frac{e^{-2}}{2} - \frac{1}{2} \right) \right) \\ &= 10 \times 10^{-3} \left(1 - \frac{e^{-2}}{2} + \frac{1}{2} \right) \\ &= 10 \times 10^{-3} \left(\frac{3}{2} - \frac{e^{-2}}{2} \right) \\ &= 14.3 \text{ mC} \end{aligned}$$

Therefore, the total charge in the device is $\boxed{14.3 \text{ mC}}$.

Step 2 of 2

(b)

Write the expression for power consumed by the device.

$$\begin{aligned} p(t) &= v(t)i(t) \\ &= [20 \sin(4t)] [10 \times 10^{-3} (1 + e^{-2t})] \\ &= 200 \sin(4t) (1 + e^{-2t}) \times 10^{-3} \text{ W} \end{aligned}$$

Calculate the power consumed by the device at $t = 1 \text{ s}$.

$$\begin{aligned} p &= 200 \sin(4 \times 1) (1 + e^{-2 \times 1}) \times 10^{-3} \\ &= 200 \sin(4) (1 + e^{-2}) \times 10^{-3} \\ &= 200 (-0.757) (1.135) \times 10^{-3} \\ &= -171.84 \text{ mW} \end{aligned}$$

Therefore, the power consumed by the device is -171.84 mW.

[Comments \(1\)](#)



Anonymous

This answer comes from being in radians, clearly, but why are we using radians?
